

In one line: App Intents (iOS 16+) exposes an app **action** to the system as plain Swift: a struct conforming to `AppIntent` with `@Parameters` and an `async perform()`. Write it **once**; Siri, Shortcuts, Spotlight, the Action button, interactive widgets, Control Center and visual intelligence all run the *same* type — no `.intentdefinition` file, usually no extension.

How it works

- **AppIntent:** static let title: LocalizedStringResource, @Parameter(title:) properties, func perform() async throws -> some IntentResult. Results: .result(), .result(value:) (chains in Shortcuts), .result(dialog:) (Siri speaks), .result(view:) (snippet).
- **Parameter types:** primitives, Date, URL..., **AppEnum** (fixed cases + caseDisplayRepresentations) or **AppEntity** (dynamic domain objects).
- **AppEntity:** id, displayRepresentation, typeDisplayRepresentation, static var defaultQuery. Its **EntityQuery:** entities(for: ids) (required — resolve saved ids), suggestedEntities() (picker), EntityStringQuery.entities(matching:) (spoken text), **26:** @ComputedProperty (getter reads your model / UserDefaults — no stored copy), @DeferredProperty (get async throws: fetched only if the system asks).
- **Asking the user:** \$room.requestValue(), requestDisambiguation, requestConfirmation, **26:** requestConfirmation(actionName: snippetIntent:) shows a snippet; requestChoice(between: dialog: view:) with Option(title: style:) returns the pick or throws on cancel. ParameterSummary = the Shortcuts-editor sentence.
- **AppShortcutsProvider:** static var appShortcuts of AppShortcut(intent: phrases:...) — in Siri, Spotlight, Shortcuts on install, zero setup. Every phrase contains \(.applicationName); max **10**; one entity/enum parameter per phrase — updateAppShortcutParameters() when its values change.
- **Where perform() runs:** background by default, app UI not loaded, about **30 s**. **26** static let supportedModes: IntentModes — .background, .foreground(.immediate / .dynamic / .deferred), combinable; in .dynamic call continueInForeground(_:alwaysConfirm:). openAppWhenRun is **deprecated in 26** ("provide supportedModes"). Widget buttons run in the widget extension; LiveActivityIntent / AudioPlaybackIntent in the app. **27** allowedExecutionTargets (IntentExecutionTargets: main app, App Intents extension, WidgetKit extension) pins the process.
- **Dependencies:** @Dependency var store: Store, registered at launch with AppDependencyManager.shared.add { store } — not singletons reached from perform().
- **Special conformances:** WidgetConfigurationIntent (17), SetFocusFilterIntent, ControlConfigurationIntent (Control Center, 18), OpenIntent (opens an entity). **26** UndoableIntent: its undoManager registers undo for the in-app action. AppIntentsPackage (17, frameworks) now also for Swift packages + static libraries.

Example

```
struct ToggleLight: AppIntent {
    static let title: LocalizedStringResource = "Toggle Light"
    static let supportedModes: IntentModes = [.background, .foreground(.dynamic)]
    @Parameter(title: "Room") var room: RoomEntity
    @Dependency var lights: LightService
    func perform() async throws -> some IntentResult & ProvidesDialog {
        let on = try await lights.toggle(room.id) // modes: iOS 26
        return .result(dialog: "\(room.name) is \(on ? "on" : "off")")
    }
}
// iOS 26: ask mid-perform with a custom snippet view
let archive = Option(title: "Archive", style: .default)
let delete = Option(title: "Delete", style: .destructive)
let pick = try await requestChoice(between: [.cancel, archive, delete],
    dialog: "Archive or delete \(album.name)?", view: AlbumCard(album))
struct RoomSnippet: SnippetIntent { // iOS 26
    @Parameter var room: RoomEntity
    func perform() async throws -> some IntentResult & ShowsSnippetView {
        .result(view: RoomCard(room)) // Button(intent: ToggleLight())...
        // RoomSnippet.reload() on change
    }
}
```

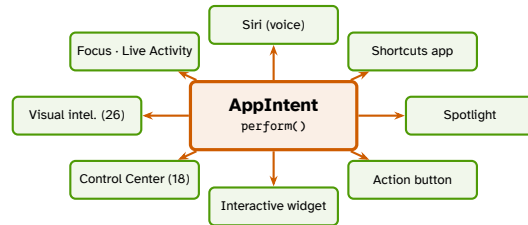
Spotlight & donation (in brief)

NSUserActivity (isEligibleForSearch/Prediction, Handoff) · Core Spotlight CSearchableItem + CSearchableIndex · IndexedEntity (18) · **donate** (IntentDonationManager.shared.donate(intent:)) after in-app use — feeds suggestions, does not expose the action.

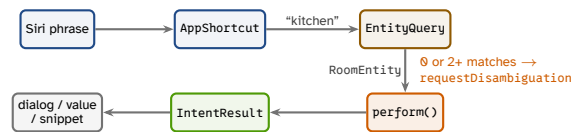
Which protocol when

need	adopt
show / edit UI inline in Siri, Spotlight	SnippetIntent (26)
open the app on an entity	OpenIntent · supportedModes (26)
configure a widget / a control	WidgetConfigurationIntent · ControlConfigurationIntent
button in a Live Activity / audio	LiveActivityIntent · AudioPlaybackIntent
undo from the app's UI	UndoableIntent (26)
be found by image search	IntentValueQuery (26)
clean up on cancel · work past 30 s	CancellableIntent (26.4) · LongRunningIntent (27)

Write once, surface everywhere



"Toggle kitchen in Lumen"



New in iOS 26 · 26.4 · 27

- **26 Interactive snippets:** SnippetIntent returns ShowsSnippetView; its buttons/toggles run other intents, then the system calls its perform() again to redraw; reload() when your data changes.
- **26 Visual intelligence:** an IntentValueQuery with values(for input: SemanticContentDescriptor) (input: pixelBuffer) returns your entities for on-screen / camera image search; @UnionValue enum when results mix entity types.
- **26.4 CancellableIntent:** withIntentCancellationHandler(operation: onCancel:) gets a reason — no progress past the 30 s limit, or the person cancelled.
- **27 LongRunningIntent** (a ProgressReportingIntent): performBackgroundTask { ... } runs past 30 s if it keeps updating progress. EntityCollection<T>: a parameter of ids only (.identifiers, resolvedEntities() when needed). SyncableEntity: id stable across devices (SyncableEntityIdentifier(local: stable:)) so Siri can hand off. **App Intents Testing** framework (AnyAppIntent, ResolvedIntentResult, IntentDefinitions).

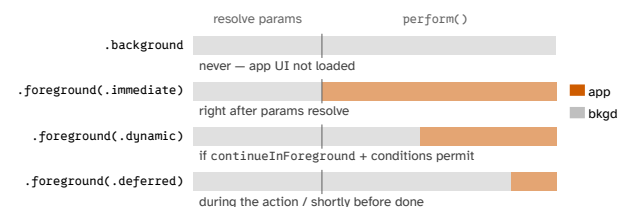
Traps

- ! Phrase without \(.applicationName) — never matches.
- ! UI from a background perform() — declare supportedModes (pre-26: openAppWhenRun, now deprecated).
- ! "App Intents replaced SiriKit" — not all: messaging, VoIP calling, media, CarPlay still use INIntent + an Intents extension.
- ! Assuming app state in perform(); entities persist by id and re-resolve via the query.
- ! Long work in perform() with no progress — cancelled at ~30 s; report progress (LongRunningIntent in 27).
- ! Expensive entity fields computed eagerly — @DeferredProperty; copied state — @ComputedProperty.

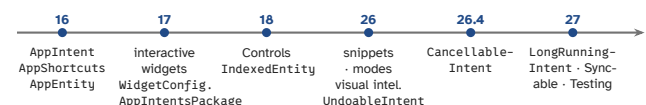
Remember

"I-E-Q-S": Intent does it, Entity names it, Query finds it, Shortcut says it — **Modes** say where it runs.

supportedModes (26) — when the app comes forward



Timeline — what arrived when



Related: widgets-live-activities (interactive widgets, Live Activities) · deep links (onOpenURL) · accessibility & localization (LocalizedStringResource) · background-execution (30 s budgets) · cloud-vs-on-device (Foundation Models tools)